**University of Florida, Department of Mathematics**

# Complex Analysis PhD Exam, September 1992

*Present the solutions with the necessary details, so the partial credit can be properly assessed. Do all 9 problems.*

**1.** Construct a conformal map which maps the vertical strip $0 < \operatorname{Re} z < 1$ onto the upper half-plane.

**2.** The function

$$f(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^2}$$

is analytic in the unit circle $|z| < 1$. Can it be continued analytically beyond the circle $|z| = 1$? Discuss the nature of the singularities (if any) on $|z| = 1$.

**3.** This problem has two parts.

(a) Show that the function

$$f(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^z}$$

is analytic in $\operatorname{Re} z > 0$.

(b) Show that it can be analytically continued to the entire complex plane.

**4.** Let $f(z)$ be analytic in the disc $|z| \le 1$. If

$$|f(z)| \le 1 \quad \text{for } |z| \le 1,$$

prove that

$$\frac{|f'(z)|}{1 - |f(z)|^2} \le \frac{1}{1 - |z|^2} \quad \text{for } |z| \le 1.$$

**5.** Let $F(z)$ be an entire function with the property that $\operatorname{Re} F(z)$ is bounded from above. Prove that $F(z)$ is constant.

**6.** Let $\Omega$ be an open connected subset of $\mathbb{C}$. Let $\{f_n\}$ be a sequence of one-to-one analytic functions on $\Omega$, and assume that $\{f_n\}$ converges uniformly on any compact subset of $\Omega$ to a function $f$. Show that $f$ is either one-to-one or a constant, and show that both possibilities can occur.

**7.** Suppose that $f$ is entire, of order $\rho$. Let $n(r)$ be the number of zeros (counting the multiplicity) of $f$ in the disk $\{z : |z| \le r\}$. Prove that

$$\limsup_{r \to \infty} \frac{\log n(r)}{\log r} \le \rho,$$

and show, by examples, that equality can happen.

**8.** The expression

$$\{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2} \left\{ \frac{f''(z)}{f'(z)} \right\}^2$$

is called the Schwarzian derivative of $f$. Prove the following:

(a) If $f(z) = a(z - z_0)^m + \cdots$, where $m$ is an integer, then

$$\{f, z\} = A(z - z_0)^{-2} + \cdots.$$

(b) Find the value of $A$.

(c) What can you say about $\{f, z\}$ at the point $z_0$ if $m = \pm 1$?

**9.** This problem has two parts.

(a) Evaluate

$$\frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \frac{x^z}{z(z+1)}\, dz$$

for $x > 1$.
Hint: Write the above integral as $\lim_{t\to\infty} I(t)$, where

$$I(t) = \frac{1}{2\pi i} \int_{2-it}^{2+it} \frac{x^z}{z(z+1)}\, dz,$$

and evaluate $I(t)$ by using the residue theorem on the rectangle $2 - it$, $2 + it$, $b + it$, and $b - it$, where $b$ is a large negative number.

(b) What happens if $0 < x < 1$?